2.1

Change in Arithmetic and Geometric Sequences

Student Notes and Guided Practice

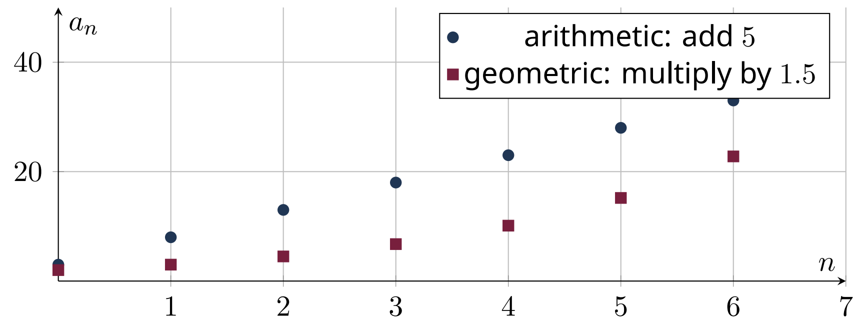
AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Learning Targets

- Distinguish arithmetic from geometric sequences.
- Write recursive and explicit sequence formulas.
- Interpret common difference, common ratio, and initial value in context.

Essential Question

How do equal additions and equal multiplications generate different kinds of change?



Concept Framework

Key Idea

Arithmetic sequences have constant difference; geometric sequences have constant ratio.

Key Idea

Sequence graphs are discrete because their domain is integer-valued.

Key Idea

Indexing may start at 0 or 1; formulas must match the chosen index.

Key Idea

A negative geometric ratio creates alternating signs.

Formula Map

Formula and Structure

$$a_n = a_0 + nd$$

Formula and Structure

$$a_n = a_1 + (n-1)d$$

Formula and Structure

$$g_n = g_0 r^n$$

Formula and Structure

$$g_n = g_1 r^{n-1}$$

Worked Examples

Worked Example 1

Problem. Write explicit and recursive formulas for $7, 12, 17, 22, \ldots$.
Solution. Arithmetic with $d = 5$: $a_n = 7 + 5(n-1)$, and $a_1 = 7,\ a_n = a_{n-1} + 5$.

Worked Example 2

Problem. Write a formula for $160, 120, 90, \ldots$.
Solution. Geometric with ratio $3/4$: $g_n = 160(3/4)^{n-1}$.

Worked Example 3

Problem. Which is larger for sufficiently large n: $5 + 20n$ or $5(1.2)^n$?
Solution. The geometric/exponential sequence eventually exceeds the arithmetic/linear sequence.

Multiple-Representation Connection

AP Reasoning

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Classify and write an explicit formula for 4,9,14,19,....

2. Classify and write an explicit formula for 81,27,9,3,....

3. Classify and write an explicit formula for 5,-10,20,-40,....

4. Classify and write an explicit formula for -7,-3,1,5,....

5. Classify and write an explicit formula for 2,6,18,54,....

6. Find the 40th term of the arithmetic sequence with $a_7 = 19$ and $a_{15} = 51$.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is ______________________________.
2. A restriction I must remember is ______________________________.
3. One graphical feature I can justify algebraically is ______________________________.

Video support: Flipped Math topic page | Original ECHS material informed by Pearson and Holt Precalculus concepts.